

One-Way Analysis of Variance

A Crash Course from First Principles

Linear Statistical Models — Lectures 11–12 (27 Aug – 31 Aug)

This is the topic the mid-semester guide is written around.

The fourth and most important of the four crash courses. Everything in Crash Courses 1–3 was preparation for this. Read §8 (the fully worked numerical problem) even if you read nothing else — the exam guide says one question **will** require a calculator.

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10 Practice, with answers

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1 Why this exists

You have r groups — r drugs, r fertilisers, r machines — and a numerical response. The question is whether the groups differ.

The concrete setting from the notes

r different drugs for reducing cholesterol. Y_{ij} is the reduction in cholesterol after one month for the j -th subject receiving the i -th drug:

$$i = 1, \dots, r \text{ (treatment index)}, \quad j = 1, \dots, n_i \text{ (replicate index)}, \quad n = \sum_i n_i.$$

The design is **balanced** if all n_i are equal; in general they need not be. Subjects are assigned to treatments *at random* — the randomisation principle — which is what lets us attribute a difference in outcome to the treatment.

With $r = 2$ this is just the two-sample t -test. The whole point of ANOVA is to handle $r > 2$ *without* doing $\binom{r}{2}$ separate t -tests (which would inflate the error rate badly). It answers one global question first: **is there any difference at all?**

2 The model, and the identifiability problem

2.1 Three equivalent ways to write it

Cell-means form (the honest one)

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, r, \quad j = 1, \dots, n_i.$$

Parameters: $\mu_1, \dots, \mu_r, \sigma^2$. This model is perfectly identifiable.

But we want to talk about *differential effects*: how much better is drug i than average? So define an overall mean μ and write $\alpha_i = \mu_i - \mu$:

Effects form (the useful one)

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

Parameters: $\mu, \alpha_1, \dots, \alpha_r, \sigma^2$ — that is $r + 1$ parameters for only r group means. **One too many.**

2.2 Why a constraint is forced on you

Replace μ by $\mu - c$ and every α_i by $\alpha_i + c$:

$$Y_{ij} = (\mu - c) + (\alpha_i + c) + \varepsilon_{ij}$$

is the *same model*. The data can never distinguish the two parameter vectors. In the language of Crash Course 2: the design matrix

$$\mathbf{X}_{n \times (r+1)} = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{1}_{n_1} & 0 & \cdots & 0 \\ \mathbf{1}_{n_2} & 0 & \mathbf{1}_{n_2} & \cdots & 0 \\ \vdots & & & \ddots & \\ \mathbf{1}_{n_r} & 0 & 0 & \cdots & \mathbf{1}_{n_r} \end{pmatrix} \quad \text{has} \quad \rho(\mathbf{X}) = r, \text{ not } r+1,$$

because column 0 is the sum of the other r . So $\mathbf{X}^T \mathbf{X}$ is singular, the model is **not identifiable**, and μ and the α_i are **not estimable** individually.

2.3 The general form of the constraint

Define the overall mean as a weighted average, $\mu = \sum_i w_i \mu_i$ with $\sum_i w_i = 1$, $0 \leq w_i \leq 1$. Substituting $\mu_i = \mu + \alpha_i$:

$$\mu = \sum_i w_i (\mu + \alpha_i) = \mu \sum_i w_i + \sum_i w_i \alpha_i = \mu + \sum_i w_i \alpha_i \quad \Rightarrow \quad \boxed{\sum_{i=1}^r w_i \alpha_i = 0.}$$

The two natural choices

$$\text{(A)} \quad w_i = \frac{1}{r} \Rightarrow \sum_i \alpha_i = 0, \quad \text{(B)} \quad w_i = \frac{n_i}{n} \Rightarrow \sum_i n_i \alpha_i = 0.$$

(B) is the “natural” constraint and the one used in the exam guide. It corresponds to $\mu = \mathbb{E}(\bar{Y}_{..}) = \frac{1}{n} \sum_i n_i \mu_i$, the mean of the whole experiment. In a balanced design ($n_i \equiv m$) the two coincide.

Common confusion: “the constraint changes the model”

It does not. The constraint is a *labelling convention* you impose to pin down one representative from a family of equivalent parameter vectors. Fitted values, SSE, the F -statistic and every estimable quantity are completely unaffected by which constraint you choose. Only the numerical values of $\hat{\mu}$ and $\hat{\alpha}_i$ change — and those are not estimable anyway.

2.4 What *is* estimable

By the criterion of Crash Course 2, $\mathcal{N}(\mathbf{X})$ is spanned by $z = (1, -1, \dots, -1)^T$, so $a^T \beta$ is estimable iff $a_0 = \sum_{i \geq 1} a_i$.

The estimable quantities of one-way ANOVA

- $\mu_i = \mu + \alpha_i$ — each group mean. **Estimable.**
- $\alpha_i - \alpha_{i'} = \mu_i - \mu_{i'}$ — pairwise differences. **Estimable.**
- $\sum_i c_i \alpha_i$ with $\sum_i c_i = 0$ — a **contrast**. **Estimable**, and

$$\sum_i c_i \alpha_i = \sum_i c_i (\mu_i - \mu) = \sum_i c_i \mu_i.$$

- μ alone, or α_i alone. **Not estimable** (without a constraint).

Common confusion: contrast vs. general linear combination

A contrast is a linear combination of the μ_i (equivalently of the α_i) whose coefficients **sum to zero**. That is the whole definition, and it is exactly the condition for estimability. $\mu_1 - \mu_2$ is a contrast ($1 - 1 = 0$); $\mu_1 + \mu_2$ is not; $\mu_2 - \frac{1}{2}(\mu_1 + \mu_3)$ is ($-\frac{1}{2} + 1 - \frac{1}{2} = 0$).

3 Least squares estimates

3.1 The direct route (fastest, use this in the exam)

Because the cell-means form has no constraint, minimise

$$G(\mu_1, \dots, \mu_r) = \sum_{i=1}^r \sum_{j=1}^{n_i} (Y_{ij} - \mu_i)^2.$$

The terms separate by i , and $\sum_j (Y_{ij} - \mu_i)^2$ is minimised at the group mean. So

$$\hat{\mu}_i = \bar{Y}_{i\cdot} = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}$$

By Gauss–Markov, since each μ_i is estimable, $\bar{Y}_{i\cdot}$ is its **BLUE**. Then under constraint (B):

$$\hat{\mu} = \bar{Y}_{..} = \frac{1}{n} \sum_i \sum_j Y_{ij}, \quad \hat{\alpha}_i = \bar{Y}_{i\cdot} - \bar{Y}_{..}$$

and indeed $\sum_i n_i \hat{\alpha}_i = \sum_i n_i \bar{Y}_{i\cdot} - n \bar{Y}_{..} = 0$: the constraint is automatically satisfied. Under the general constraint $\sum w_i \alpha_i = 0$ the BLUEs are $\hat{\mu} = \sum_i w_i \bar{Y}_{i\cdot}$ and $\hat{\alpha}_i = \bar{Y}_{i\cdot} - \hat{\mu}$.

3.2 The projection route (this is what the lectures did)

Write $\mathbf{Y} = \mu \mathbf{1}_n + \mathbf{X}\boldsymbol{\alpha} + \boldsymbol{\varepsilon}$ where $\mathbf{X}$ has the r group-indicator columns. Let $P_1 = \frac{1}{n} \mathbf{1}\mathbf{1}^T$ and $\tilde{\mathbf{X}} = (I - P_1)\mathbf{X}$. The constraint $\sum n_i \alpha_i = 0$ is exactly $\mathbf{1}^T \mathbf{X}\boldsymbol{\alpha} = 0$, so $\tilde{\mathbf{X}}\boldsymbol{\alpha} = \mathbf{X}\boldsymbol{\alpha}$, and $P_{\tilde{\mathbf{X}}_{\text{tot}}} = P_1 + P_{\tilde{\mathbf{X}}}$ by the lemma of Crash Course 3. Then

$$\begin{aligned} G(\mu, \boldsymbol{\alpha}) &= \|\mathbf{Y} - \mu \mathbf{1} - \tilde{\mathbf{X}}\boldsymbol{\alpha}\|^2 \\ &= \underbrace{\|(I - P_{\tilde{\mathbf{X}}_{\text{tot}}})\mathbf{Y}\|^2}_{\text{constant in } (\mu, \boldsymbol{\alpha})} + \underbrace{\|P_1 \mathbf{Y} - \mu \mathbf{1}\|^2}_{\text{involves only } \mu} + \underbrace{\|P_{\tilde{\mathbf{X}}} \mathbf{Y} - \mathbf{X}\boldsymbol{\alpha}\|^2}_{\text{involves only } \boldsymbol{\alpha}}, \end{aligned}$$

the three pieces being mutually orthogonal. Minimising the middle term gives $\hat{\mu} = \bar{Y}_{..}$; minimising the last gives $\hat{\boldsymbol{\alpha}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T P_{\tilde{\mathbf{X}}} \mathbf{Y}$. Since $\mathbf{X}^T \mathbf{X} = \text{diag}(n_1, \dots, n_r)$ and $\mathbf{X}^T P_{\tilde{\mathbf{X}}} \mathbf{Y} = \mathbf{X}^T \mathbf{Y} - \bar{Y}_{..} \mathbf{X}^T \mathbf{1} = (\sum_j Y_{1j} - n_1 \bar{Y}_{..}, \dots)^T$, we recover $\hat{\alpha}_i = \bar{Y}_{i\cdot} - \bar{Y}_{..}$. ■

3.3 The estimate of σ^2

$$\hat{\sigma}^2 = \text{MSE} = \frac{\text{SSE}}{n - r} = \frac{\sum_{i=1}^r (n_i - 1) s_i^2}{n - r}, \quad s_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i\cdot})^2.$$

This is the **pooled sample variance** — a weighted average of the r within-group variances, with weights equal to their degrees of freedom. It is unbiased for σ^2 . Note $\sum_i (n_i - 1) = n - r$.

4 The variance decomposition

4.1 Definitions and the identity

Sums of squares

$$\begin{aligned} \text{SSTO} &= \sum_i \sum_j (Y_{ij} - \bar{Y}_{..})^2 && \text{total variability} \\ \text{SSTR} &= \sum_{i=1}^r n_i (\bar{Y}_{i.} - \bar{Y}_{..})^2 && \text{between groups (BG)} \\ \text{SSE} &= \sum_i \sum_j (Y_{ij} - \bar{Y}_{i.})^2 && \text{within groups (WG)} \end{aligned}$$

$$\boxed{\text{SSTO} = \text{SSTR} + \text{SSE}}$$

Proof. (algebraic) Insert and split: $Y_{ij} - \bar{Y}_{..} = (Y_{ij} - \bar{Y}_{i.}) + (\bar{Y}_{i.} - \bar{Y}_{..})$. Squaring and summing, the cross term is

$$2 \sum_i (\bar{Y}_{i.} - \bar{Y}_{..}) \underbrace{\sum_j (Y_{ij} - \bar{Y}_{i.})}_{=0} = 0. \quad \square$$

Proof. (geometric — this is the lecture version) With P_1 and P_X the projections onto the constant vector and onto the group-indicator space,

$$(I - P_1)\mathbf{Y} = \underbrace{(I - P_X)\mathbf{Y}}_{\perp} + \underbrace{(P_X - P_1)\mathbf{Y}}_{\perp},$$

the two being orthogonal because $\mathcal{C}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$. Pythagoras gives $\mathbf{Y}^T(I - P_1)\mathbf{Y} = \mathbf{Y}^T(I - P_X)\mathbf{Y} + \mathbf{Y}^T(P_X - P_1)\mathbf{Y}$, which is $\text{SSTO} = \text{SSE} + \text{SSTR}$. ■

The picture: two scenes

Scene 1: groups clearly differ

Points within each group are tightly bunched; the group centres are far apart.

SSE (WG) $\ll$ SSTR (BG)

F large $\Rightarrow$ **reject** H_0

Scene 2: groups indistinguishable

Points within each group are spread out; the group centres are close together.

SSTR (BG) is small relative to SSE (WG)

F near 1 $\Rightarrow$ **do not reject**

The F -statistic is nothing but the formal version of “how big is the spread *between* groups compared to the spread *within* groups?”

4.2 Degrees of freedom

$$\underbrace{n-1}_{\rho(I-P_1)} = \underbrace{r-1}_{\rho(P_X-P_1)} + \underbrace{n-r}_{\rho(I-P_X)}.$$

The ranks add up to $n-1$, so **Fisher–Cochran** applies (after removing the mean) and gives: SSTR and SSE are *independent* chi-squares.

4.3 Expected mean squares

The table that explains why F works

$$\mathbb{E}(\text{MSE}) = \sigma^2, \quad \mathbb{E}(\text{MSTR}) = \sigma^2 + \frac{\sum_{i=1}^r n_i (\mu_i - \mu)^2}{r-1} = \sigma^2 + \frac{\sum_i n_i \alpha_i^2}{r-1}.$$

Under $H_0 : \alpha_1 = \dots = \alpha_r = 0$ the second term vanishes and *both* mean squares estimate σ^2 , so $F = \text{MSTR}/\text{MSE}$ should hover around 1. Under H_1 the numerator is inflated by $\sum n_i \alpha_i^2$, so F tends to be large. **That is why you reject for large F and never for small F : the test is one-sided in F even though the alternative is two-sided in the α_i .**

5 The ANOVA table and the F -test

The table — reproduce this from memory

Source of variation	d.f.	SS	MS = $\frac{\text{SS}}{\text{d.f.}}$	$\mathbb{E}(\text{MS})$
Between treatments	$r-1$	SSTR	$\text{MSTR} = \frac{\text{SSTR}}{r-1}$	$\sigma^2 + \frac{\sum_i n_i (\mu_i - \mu)^2}{r-1}$
Within (Error)	$n-r$	SSE	$\text{MSE} = \frac{\text{SSE}}{n-r}$	σ^2
Total	$n-1$	SSTO		

$$H_0 : \alpha_1 = \dots = \alpha_r = 0 \quad (\text{equivalently } \mu_1 = \dots = \mu_r), \quad F = \frac{\text{MSTR}}{\text{MSE}} = \frac{\text{SSTR}/(r-1)}{\text{SSE}/(n-r)} \stackrel{H_0}{\sim} F_{r-1, n-r}.$$

Reject H_0 at level α if $F > F_{r-1, n-r}^{1-\alpha}$.

Under H_1 , F has a **non-central** $F_{r-1, n-r}(\lambda/2)$ distribution with $\lambda = \frac{1}{\sigma^2} \sum_i n_i \alpha_i^2$; the power increases to 1 as $\lambda \rightarrow \infty$.

Common confusion: $n-r$, not $n-r-1$

The model $\mu + \alpha_i$ has $r+1$ parameters but the design matrix has rank r . The error d.f. is $n - \rho(\mathbf{X}) = n - r$. Every t and F in the question uses this. Writing $n-r-1$ poisons the whole answer.

6 Confidence intervals and contrasts

Everything here uses one distributional fact: $\bar{Y}_{i\cdot} \sim \mathcal{N}(\mu_i, \sigma^2/n_i)$ independently across i , independent of MSE, and $(n-r)\text{MSE}/\sigma^2 \sim \chi_{n-r}^2$. So any (estimable) linear combination, standardised by its estimated standard error, is t_{n-r} .

6.1 A single group mean

$$\hat{\mu}_i = \bar{Y}_{i\cdot}, \quad \text{Var}(\hat{\mu}_i) = \frac{\sigma^2}{n_i}, \quad \boxed{\bar{Y}_{i\cdot} \pm t_{n-r}^{1-\alpha/2} \sqrt{\frac{\text{MSE}}{n_i}}}$$

6.2 A pairwise difference

$$\widehat{\mu_i - \mu_{i'}} = \bar{Y}_{i\cdot} - \bar{Y}_{i'\cdot}, \quad \boxed{(\bar{Y}_{i\cdot} - \bar{Y}_{i'\cdot}) \pm t_{n-r}^{1-\alpha/2} \sqrt{\text{MSE} \left(\frac{1}{n_i} + \frac{1}{n_{i'}} \right)}}$$

Note this is exactly the two-sample t interval, except that σ^2 is estimated by pooling *all* r groups, giving $n - r$ degrees of freedom instead of $n_i + n_{i'} - 2$. That extra precision is the payoff of the ANOVA framework.

6.3 A general contrast — the master formula

Contrast: estimate, variance, interval

Let $\theta = \sum_{i=1}^r c_i \mu_i$ with $\sum_i c_i = 0$ (equivalently $\theta = \sum_i c_i \alpha_i$).

$$\hat{\theta} = \sum_{i=1}^r c_i \bar{Y}_{i.} \quad \text{is the BLUE,} \quad \text{Var}(\hat{\theta}) = \sigma^2 \sum_{i=1}^r \frac{c_i^2}{n_i},$$

$$\sum_i c_i \bar{Y}_{i.} \pm t_{n-r}^{1-\alpha/2} \sqrt{\text{MSE}} \sqrt{\sum_{i=1}^r \frac{c_i^2}{n_i}}$$

and the test of $H_0 : \theta = 0$ rejects when $|\hat{\theta} / \sqrt{\text{MSE} \sum c_i^2 / n_i}| > t_{n-r}^{1-\alpha/2}$.

Why $\hat{\theta}$ is the BLUE. θ is estimable (the coefficients sum to zero), $\bar{Y}_{i.} = \hat{\mu}_i$ is the LS estimate, so $\sum c_i \bar{Y}_{i.}$ is “ $a^T \hat{\beta}$ ” and the Gauss–Markov theorem applies directly.

Why the variance is that. The $\bar{Y}_{i.}$ are independent with variances σ^2/n_i , so $\text{Var}(\sum c_i \bar{Y}_{i.}) = \sum c_i^2 \sigma^2 / n_i$.

Common confusion: $\sqrt{\text{MSE}} \cdot \sqrt{\sum c_i^2 / n_i}$ — one square root or two?

It is a single square root of the product: $\sqrt{\text{MSE} \sum c_i^2 / n_i}$. The exam guide writes it as $\sqrt{\text{MSE}} \sqrt{\sum c_i^2 / n_i}$, which is the same thing. What you must *not* do is forget the $1/n_i$ weights — with unequal group sizes that is the most common numerical error in the whole topic.

7 The LS decomposition under the natural constraint

The exam guide devotes a page to this argument, so here it is in full. We minimise $G(\mu, \alpha_1, \dots, \alpha_r) = \sum_i \sum_j (Y_{ij} - \mu - \alpha_i)^2$ subject to $\sum_i n_i \alpha_i = 0$ ($\star$).

Insert $\hat{\mu} = \bar{Y}_{..}$ and $\hat{\alpha}_i = \bar{Y}_{i.} - \bar{Y}_{..}$ (so $\hat{\mu} + \hat{\alpha}_i = \bar{Y}_{i.}$) and split the summand three ways:

$$Y_{ij} - \mu - \alpha_i = \underbrace{(Y_{ij} - (\hat{\mu} + \hat{\alpha}_i))}_{= Y_{ij} - \bar{Y}_{i.}} - \underbrace{(\mu - \hat{\mu})}_{\text{constant in } i, j} - \underbrace{(\alpha_i - \hat{\alpha}_i)}_{\text{constant in } j}.$$

Squaring and summing, all three cross terms vanish:

- $\sum_i \sum_j (Y_{ij} - \bar{Y}_{i.})(\mu - \hat{\mu}) = (\mu - \hat{\mu}) \sum_i \sum_j (Y_{ij} - \bar{Y}_{i.}) = 0$;
- $\sum_i (\alpha_i - \hat{\alpha}_i) \sum_j (Y_{ij} - \bar{Y}_{i.}) = 0$ likewise;
- $\sum_i n_i (\mu - \hat{\mu})(\alpha_i - \hat{\alpha}_i) = (\mu - \hat{\mu})(\sum_i n_i \alpha_i - \sum_i n_i \hat{\alpha}_i) = 0$ — **and this is precisely where the constraint ($\star$) is used**, both for α and for $\hat{\alpha}$.

Therefore

$$G(\mu, \alpha) = \underbrace{\sum_i \sum_j (Y_{ij} - \bar{Y}_{i.})^2}_{= \text{SSE, free of } \mu, \alpha} + n(\mu - \hat{\mu})^2 + \sum_i n_i (\alpha_i - \hat{\alpha}_i)^2.$$

Each of the last two terms is a non-negative quadratic minimised at $\mu = \hat{\mu}$, $\alpha_i = \hat{\alpha}_i$ — and that choice satisfies $(\star)$, so it is admissible. Setting $\mu = \hat{\mu}$ and all $\alpha_i = 0$ instead (also admissible) and comparing gives

$$\sum_i \sum_j (Y_{ij} - \bar{Y}_{..})^2 = \sum_i \sum_j (Y_{ij} - \bar{Y}_{i.})^2 + \sum_i n_i \hat{\alpha}_i^2,$$

i.e. $\text{SSTO} = \text{SSE} + \text{SSTR}$ once more. ■

Why this argument is worth knowing

It proves *simultaneously* that (a) the LSEs are the sample means, (b) they satisfy the constraint automatically, and (c) the ANOVA identity holds — all from one completed square. If the exam asks you to “derive the LSE under the natural identifiability constraint”, this is the expected answer.

8 The fully worked numerical problem

The guide is explicit: *one numerical problem, bring a calculator*. It also tells you the likely form: **you are given $\sum_i \sum_j Y_{ij}^2$ and the group totals $T_i = \sum_j Y_{ij}$, and asked to find SSTO, SSTR, SSE and everything downstream**. So learn the shortcut formulas, not the definitional ones.

8.1 The shortcut formulas

The three-line computational recipe

Let $T_i = \sum_j Y_{ij}$ (group totals), $G = \sum_i T_i$ (grand total), $n = \sum_i n_i$. Compute the **correction factor** once:

$$\text{CF} = \frac{G^2}{n}.$$

Then

$$\text{SSTO} = \sum_i \sum_j Y_{ij}^2 - \text{CF}, \quad \text{SSTR} = \sum_{i=1}^r \frac{T_i^2}{n_i} - \text{CF}, \quad \text{SSE} = \text{SSTO} - \text{SSTR}.$$

Never compute SSE from its definition — get it by subtraction. It is faster and it double-checks the identity.

Why: $\sum_i \sum_j (Y_{ij} - \bar{Y}_{..})^2 = \sum_i \sum_j Y_{ij}^2 - n\bar{Y}_{..}^2$ and $n\bar{Y}_{..}^2 = G^2/n$; similarly $\sum_i n_i(\bar{Y}_{i.} - \bar{Y}_{..})^2 = \sum_i n_i \bar{Y}_{i.}^2 - n\bar{Y}_{..}^2 = \sum_i T_i^2/n_i - G^2/n$.

8.2 The problem

Worked example

Three drugs are compared for cholesterol reduction. The observed reductions (mg/dL) are

Drug A ($n_1 = 4$):	12, 15, 11, 14
Drug B ($n_2 = 5$):	18, 20, 17, 19, 21
Drug C ($n_3 = 3$):	9, 13, 11

- Construct the ANOVA table and test $H_0 : \mu_1 = \mu_2 = \mu_3$ at the 5% level.
- Give a 95% confidence interval for μ_2 .

- (c) Give a 95% confidence interval for $\mu_1 - \mu_3$ and interpret.
 (d) Give a 95% confidence interval for the contrast $\mu_2 - \frac{1}{2}(\mu_1 + \mu_3)$.
 (e) Report $\hat{\mu}$ and $\hat{\alpha}_i$ under the constraint $\sum n_i \alpha_i = 0$.

Given: $F_{2,9}^{0.95} = 4.26$, $t_9^{0.975} = 2.262$.

8.3 Step 1 — the table of totals (do this first, always)

Group	n_i	$T_i = \sum_j Y_{ij}$	$\bar{Y}_{i\cdot} = T_i/n_i$	T_i^2/n_i	$\sum_j Y_{ij}^2$
A	4	52	13	$52^2/4 = 676$	$144+225+121+196 = 686$
B	5	95	19	$95^2/5 = 1805$	$324+400+289+361+441 = 1815$
C	3	33	11	$33^2/3 = 363$	$81+169+121 = 371$
Total	$n = 12$	$G = 180$	$\bar{Y}_{..} = 15$	2844	2872

8.4 Step 2 — the sums of squares

$$CF = \frac{G^2}{n} = \frac{180^2}{12} = \frac{32400}{12} = 2700.$$

$$SSTO = 2872 - 2700 = \mathbf{172}, \quad SSTR = 2844 - 2700 = \mathbf{144}, \quad SSE = 172 - 144 = \mathbf{28}.$$

8.5 Step 3 — the ANOVA table

Source	d.f.	SS	MS	F
Between treatments	$r - 1 = 2$	144	$144/2 = 72$	$72/3.1111 = \mathbf{23.14}$
Error (within)	$n - r = 9$	28	$28/9 = 3.1111$	
Total	11	172		

Conclusion (a). $F = 23.14 > F_{2,9}^{0.95} = 4.26$, so **reject** H_0 at the 5% level. There is strong evidence that the three drugs differ in mean cholesterol reduction. (In fact $p \approx 0.00028$.)

8.6 Step 4 — the intervals

Throughout, $MSE = 3.1111$, $\sqrt{MSE} = 1.7638$, $t_9^{0.975} = 2.262$.

(b) **CI for μ_2 .**

$$\bar{Y}_{2\cdot} \pm t \sqrt{\frac{MSE}{n_2}} = 19 \pm 2.262 \sqrt{\frac{3.1111}{5}} = 19 \pm 2.262(0.7888) = 19 \pm 1.784 = \boxed{(17.22, 20.78)}.$$

(c) **CI for $\mu_1 - \mu_3$.** Here $c = (1, 0, -1)$, so $\sum c_i^2/n_i = \frac{1}{4} + \frac{1}{3} = 0.58333$.

$$\hat{\theta} = 13 - 11 = 2, \quad se = \sqrt{3.1111 \times 0.58333} = \sqrt{1.8148} = 1.3472,$$

$$2 \pm 2.262(1.3472) = 2 \pm 3.047 = \boxed{(-1.05, 5.05)}.$$

Interpretation: the interval *contains zero*, so at the 5% level we cannot conclude that drugs A and C differ — even though the overall F -test was highly significant. The global test says *somebody* differs; it does not say every pair does.

(d) **CI for $\mu_2 - \frac{1}{2}(\mu_1 + \mu_3)$.** Coefficients $c = (-\frac{1}{2}, 1, -\frac{1}{2})$; check $\sum c_i = 0$. ✓

$$\sum \frac{c_i^2}{n_i} = \frac{0.25}{4} + \frac{1}{5} + \frac{0.25}{3} = 0.0625 + 0.2 + 0.08333 = 0.34583,$$

$$\hat{\theta} = 19 - \frac{1}{2}(13 + 11) = 19 - 12 = 7, \quad \text{se} = \sqrt{3.1111 \times 0.34583} = \sqrt{1.0759} = 1.0373,$$

$$7 \pm 2.262(1.0373) = 7 \pm 2.347 = \boxed{(4.65, 9.35)}.$$

Zero is excluded, and comfortably: drug B is significantly better than the average of A and C .

(e) **Effects.** $\hat{\mu} = \bar{Y}_{..} = 15$ and

$$\hat{\alpha}_1 = 13 - 15 = -2, \quad \hat{\alpha}_2 = 19 - 15 = 4, \quad \hat{\alpha}_3 = 11 - 15 = -4.$$

Check the constraint: $\sum n_i \hat{\alpha}_i = 4(-2) + 5(4) + 3(-4) = -8 + 20 - 12 = 0$. ✓

Calculator discipline — do these checks, they cost seconds

1. $\sum_i T_i = G$. (Here $52 + 95 + 33 = 180$. ✓)
2. $\sum_i n_i = n$, and $\sum_i (\text{group } \sum Y^2) = \text{overall } \sum \sum Y^2$.
3. $\text{SSTR} + \text{SSE} = \text{SSTO}$ and $(r - 1) + (n - r) = n - 1$.
4. $\sum_i n_i \hat{\alpha}_i = 0$.
5. All three SS are **non-negative**. A negative SS means an arithmetic slip — usually the correction factor.
6. Keep MSE to 4 decimals; rounding it to 3.1 early shifts F by about 1% and can flip a borderline decision.

9 Loose ends worth five minutes

9.1 The two-sample t -test as the $r = 2$ case

With $r = 2$,

$$T_{12} = \frac{\bar{Y}_{1.} - \bar{Y}_{2.}}{\hat{\sigma} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n-2}, \quad \text{and} \quad T_{12}^2 = F \sim F_{1, n-2}.$$

Under H_1 the non-centrality parameter is $\frac{\mu_1 - \mu_2}{\sigma \sqrt{1/n_1 + 1/n_2}}$.

9.2 Why not just do all pairwise t -tests?

With $r = 5$ there are 10 pairs; at level 0.05 each, the chance of at least one false rejection is far above 0.05. The F -test controls one global error rate. If it rejects, *then* you look at contrasts — ideally with a Bonferroni correction (divide α by the number of comparisons; see Crash Course 1 §6.5).

9.3 The BLUE statement in the guide's language

The guide records: for $\theta = \sum c_i \mu_i$ with $\sum c_i = 0$, the BLUE is $\sum c_i \bar{Y}_{i.}$, with $\text{Var}(\hat{\theta}) = \sigma^2 \sum c_i^2 / m_i$ and the $100(1 - \gamma)\%$ interval $\sum c_i \bar{Y}_{i.} \pm t_{n_T - r}^{1 - \gamma/2} \sqrt{\text{MSE}} \sqrt{\sum c_i^2 / m_i}$. That is §6.3 above with m_i for n_i and n_T for n . Same formula; be ready for either notation.

10 Practice, with answers

- P1.** Four fertilisers, $n_i = 6$ each ($n = 24$). You are given $\sum \sum Y_{ij}^2 = 5120$, $G = 336$, and group totals $T = (72, 96, 84, 84)$. Build the ANOVA table and test at 5% given $F_{3,20}^{0.95} = 3.10$.
- P2.** In P1, give a 95% CI for $\mu_2 - \mu_1$ ($t_{20}^{0.975} = 2.086$).
- P3.** Show that $SSE = \sum_i (n_i - 1)s_i^2$, and hence that MSE is a weighted average of the within-group sample variances.
- P4.** Prove $\mathbb{E}(\text{MSTR}) = \sigma^2 + \frac{1}{r-1} \sum_i n_i \alpha_i^2$.
- P5.** Why is $\hat{\alpha}_1 = \bar{Y}_1 - \bar{Y}..$ *not* the BLUE of α_1 , strictly speaking? What is the precise correct statement?

Answers

P1. $CF = 336^2/24 = 112896/24 = 4704$. Check $\sum T_i = 72 + 96 + 84 + 84 = 336$ ✓. $\sum T_i^2/n_i = (5184 + 9216 + 7056 + 7056)/6 = 28512/6 = 4752$.

$$SSTO = 5120 - 4704 = 416, \quad SSTR = 4752 - 4704 = 48, \quad SSE = 416 - 48 = 368.$$

d.f.: 3, 20, 23. $\text{MSTR} = 48/3 = 16$, $\text{MSE} = 368/20 = 18.4$, $F = 16/18.4 = 0.87$. Since $0.87 < 3.10$, **do not reject**: no evidence that the fertilisers differ. (Note $F < 1$ — perfectly possible, and always a signal that the between-group spread is smaller than chance would typically produce.)

P2. $\bar{Y}_2 - \bar{Y}_1 = 16 - 12 = 4$; $\sum c_i^2/n_i = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$; $\text{se} = \sqrt{18.4/3} = \sqrt{6.1333} = 2.4766$. Interval $4 \pm 2.086(2.4766) = 4 \pm 5.166 = (-1.17, 9.17)$ — contains zero, consistent with the non-significant F .

P3. By definition $s_i^2 = \frac{1}{n_i-1} \sum_j (Y_{ij} - \bar{Y}_{i.})^2$, so $(n_i - 1)s_i^2 = \sum_j (Y_{ij} - \bar{Y}_{i.})^2$; summing over i gives SSE. Hence $\text{MSE} = \sum_i (n_i - 1)s_i^2 / \sum_i (n_i - 1)$, a weighted average with weights $n_i - 1$.

P4. Write $\bar{Y}_{i.} = \mu + \alpha_i + \bar{\varepsilon}_{i.}$ and $\bar{Y}.. = \mu + \bar{\varepsilon}..$ (using $\sum n_i \alpha_i = 0$). Then $\bar{Y}_{i.} - \bar{Y}.. = \alpha_i + (\bar{\varepsilon}_{i.} - \bar{\varepsilon}..)$ and

$$\mathbb{E}(\text{SSTR}) = \sum_i n_i \left[\alpha_i^2 + \mathbb{E}(\bar{\varepsilon}_{i.} - \bar{\varepsilon}..)^2 \right] = \sum_i n_i \alpha_i^2 + \sigma^2(r-1),$$

because $\sum_i n_i \mathbb{E}(\bar{\varepsilon}_{i.} - \bar{\varepsilon}..)^2 = \sum_i n_i \left(\frac{1}{n_i} + \frac{1}{n} \right) \sigma^2 = (r-1)\sigma^2$ (the cross terms vanish since $\mathbb{E}[\bar{\varepsilon}_{i.}\bar{\varepsilon}..] = \sigma^2/n$). Dividing by $r-1$ gives the result.

P5. α_1 is *not estimable* in the unconstrained model — no unbiased estimator of it exists, so it has no BLUE. The correct statement is: *under the identifiability constraint* $\sum_i n_i \alpha_i = 0$, the parameter $\alpha_1 = \mu_1 - \sum_i (n_i/n)\mu_i$ is a contrast (its coefficients $1 - n_1/n, -n_2/n, \dots$ sum to zero), and *as such* it is estimable with BLUE $\bar{Y}_1 - \bar{Y}..$. The constraint is what turns α_1 into a well-defined estimable function.

Companions: Simple Linear Regression; Matrix Algebra and the Gauss–Markov Framework; Quadratic Forms, Nested Models and Testing.